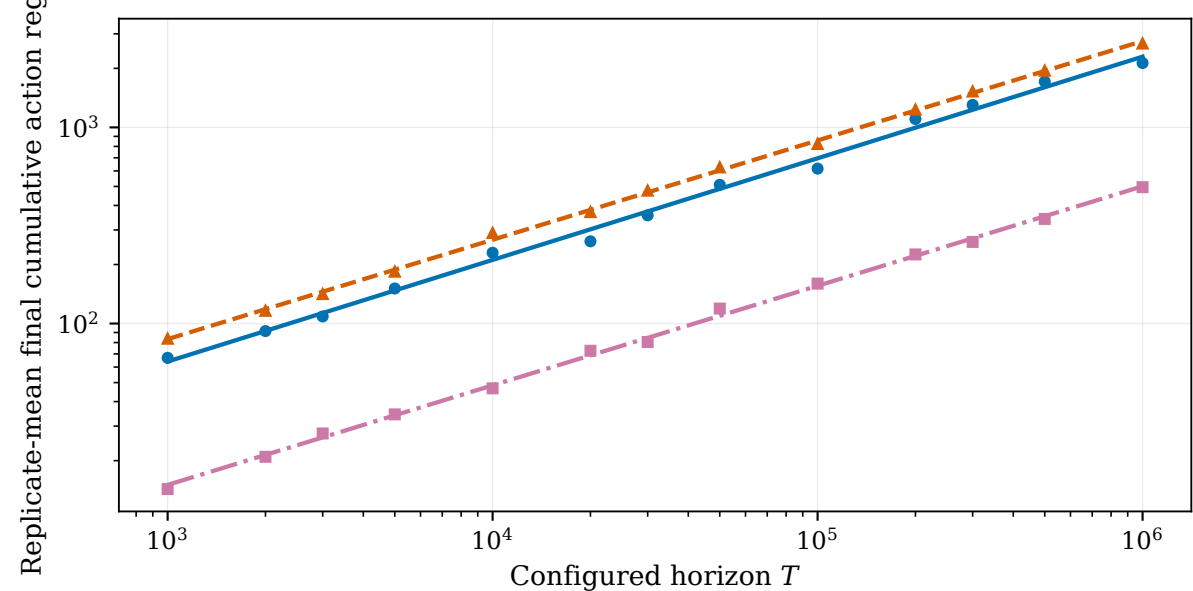


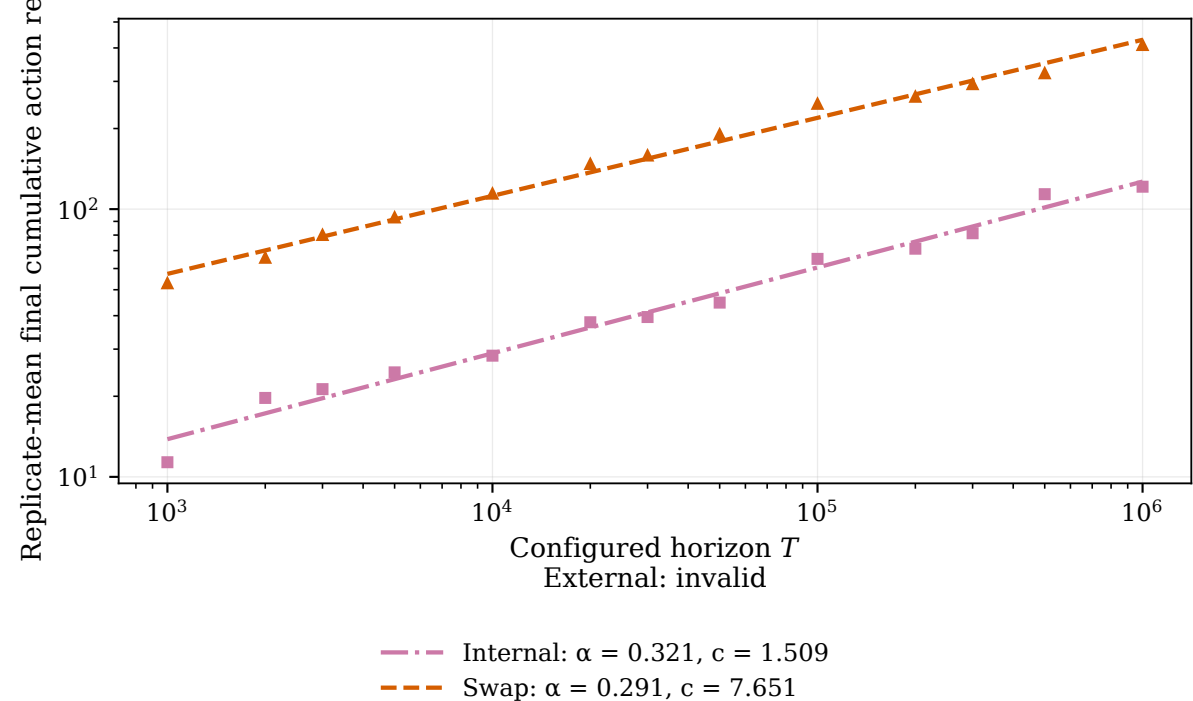
Experiment Information

Figure: Horizon scaling
Compare: Horizons
Environment: Lazy random walk
Actions: 10
Feedback: Full information
Algorithm: BM-Hedge
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.00212853, SD = 0.000752986, SE = 0.000217368
Internal R/T: mean = 0.000495717, SD = 8.48667e-05, SE = 2.44989e-05
Swap R/T: mean = 0.00267633, SD = 0.000439674, SE = 0.000126923
External: $\alpha = 0.518$, $c = 1.787$, $R^2 = 0.995$
Internal: $\alpha = 0.507$, $c = 0.452$, $R^2 = 0.999$
Swap: $\alpha = 0.506$, $c = 2.519$, $R^2 = 0.999$



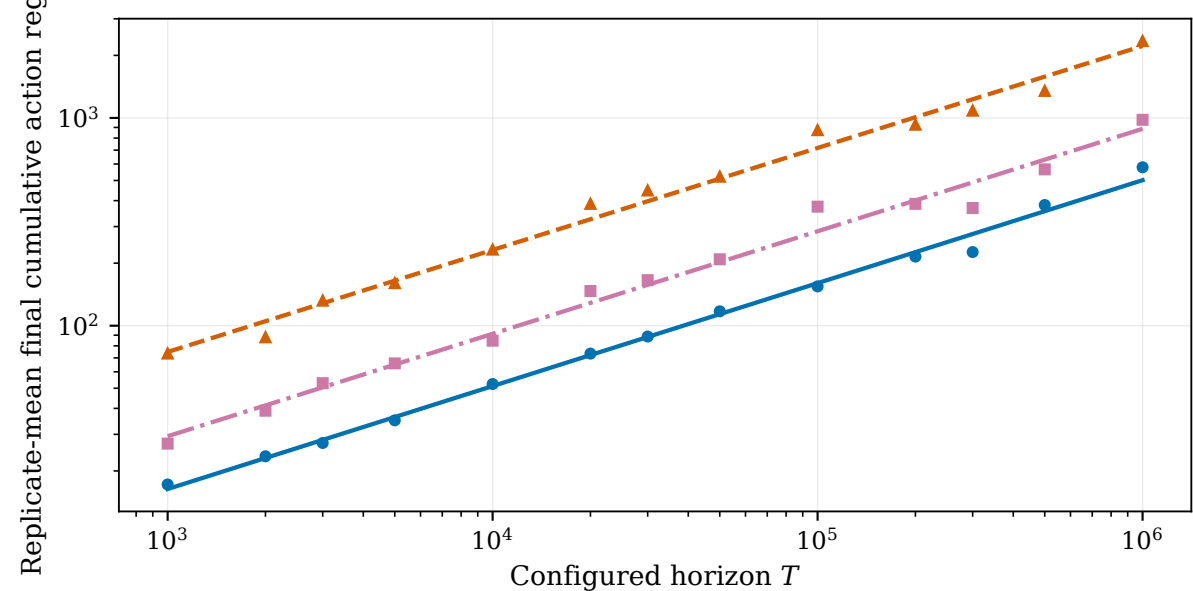
Experiment Information

Figure: Horizon scaling
Compare: Horizons
Environment: Lazy random walk
Actions: 10
Feedback: Full information
Algorithm: BM-OptHedge
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = -0.00220859, SD = 0.00143739, SE = 0.00041494
Internal R/T: mean = 0.000121275, SD = 4.25657e-05, SE = 1.22877e-05
Swap R/T: mean = 0.000410117, SD = 6.23952e-05, SE = 1.80119e-05
External: invalid
Internal: $\alpha = 0.321$, $c = 1.509$, $R^2 = 0.983$
Swap: $\alpha = 0.291$, $c = 7.651$, $R^2 = 0.991$



Experiment Information

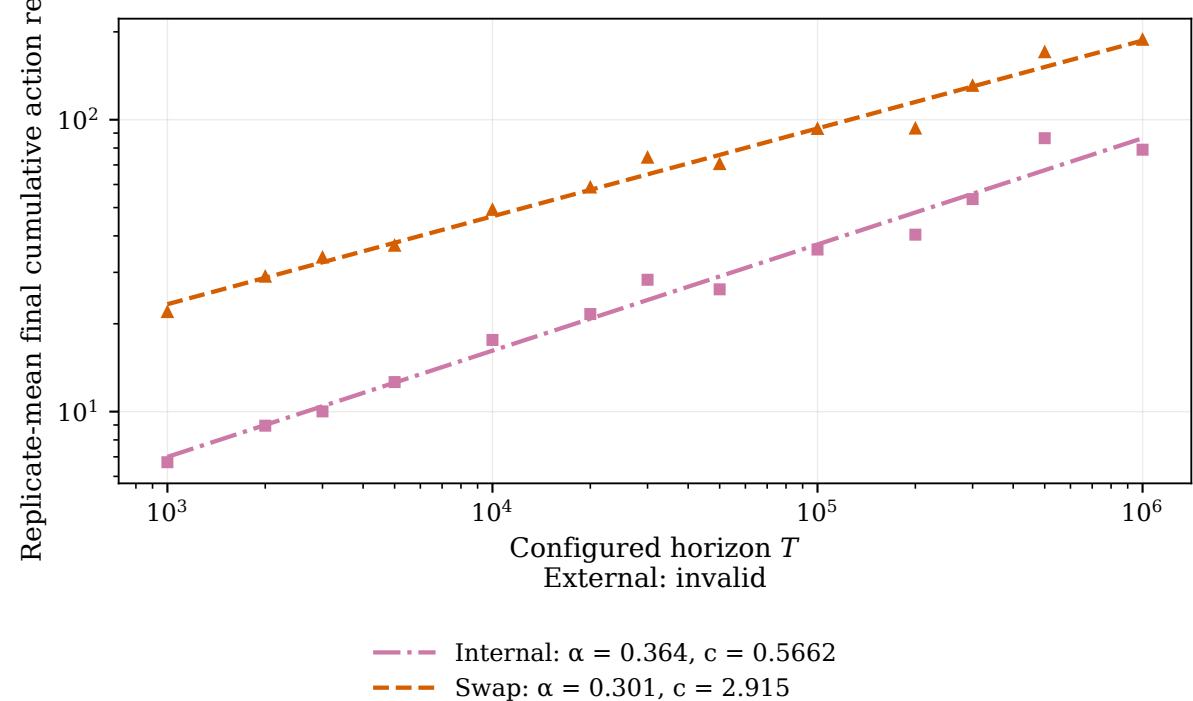
Figure: Horizon scaling
Compare: Horizons
Environment: Lazy random walk
Actions: 10
Feedback: Full information
Algorithm: Hedge
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.000578758, SD = 0.0001043, SE = 3.01089e-05
Internal R/T: mean = 0.000979592, SD = 0.000657893, SE = 0.000189917
Swap R/T: mean = 0.00234382, SD = 0.00112066, SE = 0.000323507
External: $\alpha = 0.495$, $c = 0.5339$, $R^2 = 0.995$
Internal: $\alpha = 0.493$, $c = 0.973$, $R^2 = 0.985$
Swap: $\alpha = 0.491$, $c = 2.518$, $R^2 = 0.989$



- External: $\alpha = 0.495, c = 0.5339$
- · - Internal: $\alpha = 0.493, c = 0.973$
- - Swap: $\alpha = 0.491, c = 2.518$

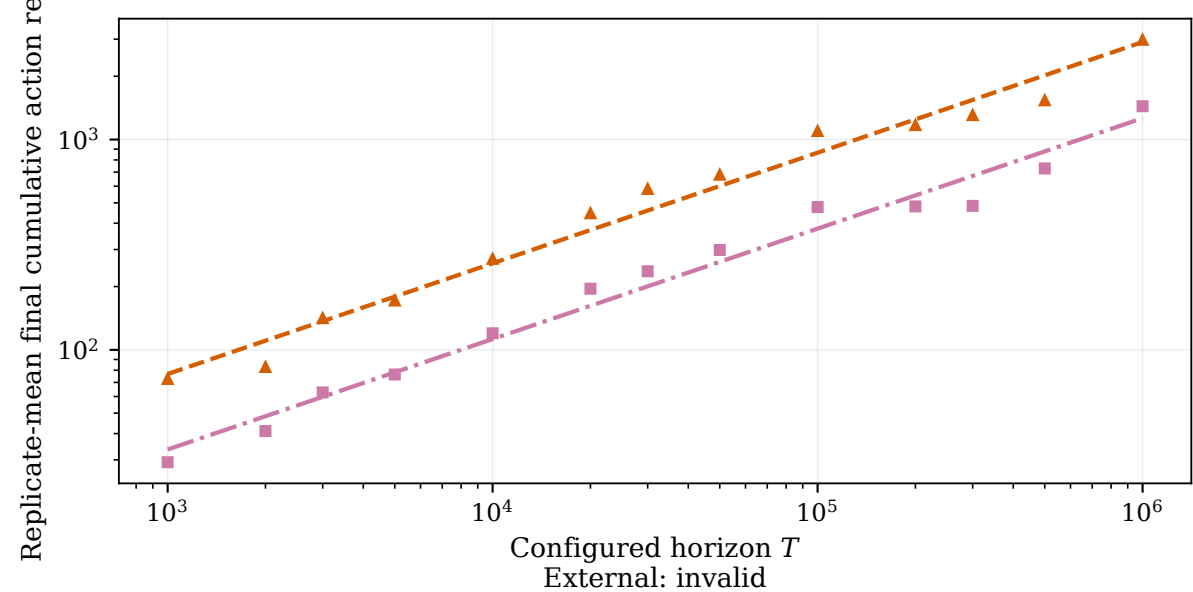
Experiment Information

Figure: Horizon scaling
Compare: Horizons
Environment: Lazy random walk
Actions: 10
Feedback: Full information
Algorithm: Ito-Hedge
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = -0.00281221, SD = 0.00150603, SE = 0.000434753
Internal R/T: mean = 7.895e-05, SD = 4.12671e-05, SE = 1.19128e-05
Swap R/T: mean = 0.000187867, SD = 9.92245e-05, SE = 2.86437e-05
External: invalid
Internal: $\alpha = 0.364$, $c = 0.5662$, $R^2 = 0.981$
Swap: $\alpha = 0.301$, $c = 2.915$, $R^2 = 0.984$



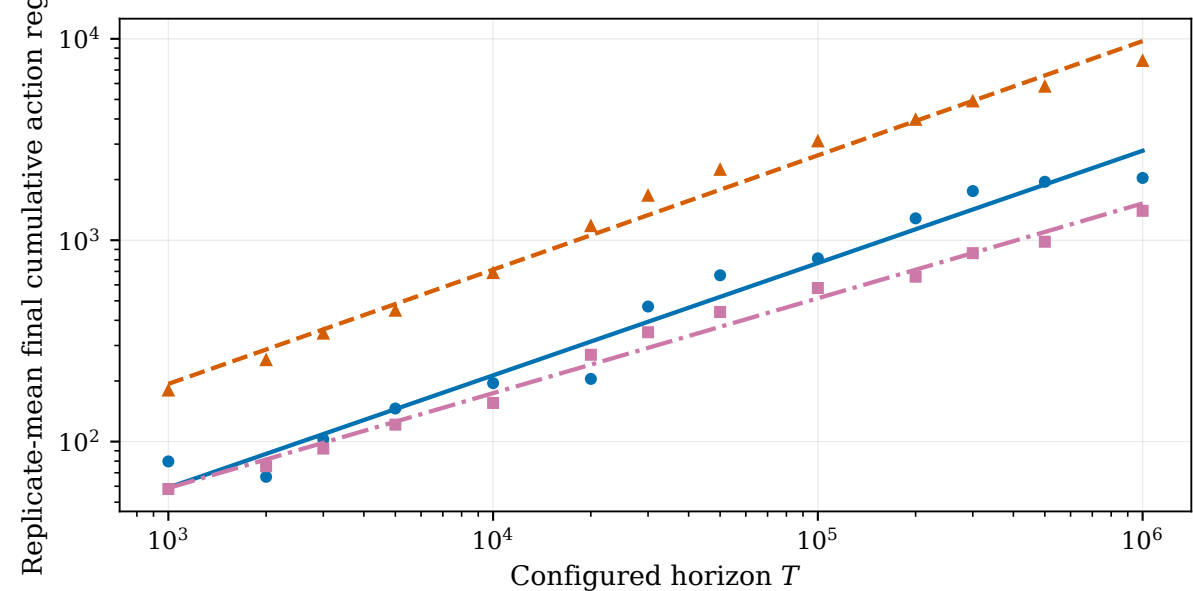
Experiment Information

Figure: Horizon scaling
Compare: Horizons
Environment: Lazy random walk
Actions: 10
Feedback: Full information
Algorithm: OptHedge
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = -5.24417e-05, SD = 4.36894e-05, SE = 1.2612e-05
Internal R/T: mean = 0.00144024, SD = 0.000797076, SE = 0.000230096
Swap R/T: mean = 0.00298063, SD = 0.00159729, SE = 0.000461098
External: invalid
Internal: $\alpha = 0.525$, $c = 0.8932$, $R^2 = 0.978$
Swap: $\alpha = 0.526$, $c = 2.028$, $R^2 = 0.979$



Experiment Information

Figure: Horizon scaling
Compare: Horizons
Environment: Lazy random walk
Actions: 10
Feedback: Full information
Algorithm: RM
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.00203804, SD = 0.00230171, SE = 0.000664446
Internal R/T: mean = 0.0013993, SD = 0.000615364, SE = 0.00017764
Swap R/T: mean = 0.00776705, SD = 0.00323116, SE = 0.000932755
External: $\alpha = 0.558$, $c = 1.254$, $R^2 = 0.968$
Internal: $\alpha = 0.471$, $c = 2.27$, $R^2 = 0.990$
Swap: $\alpha = 0.567$, $c = 3.838$, $R^2 = 0.988$



- External: $\alpha = 0.558, c = 1.254$
- Internal: $\alpha = 0.471, c = 2.27$
- Swap: $\alpha = 0.567, c = 3.838$

Experiment Information

Figure: Horizon scaling
Compare: Horizons
Environment: Lazy random walk
Actions: 10
Feedback: Full information
Algorithm: SRM
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = -0.00364579, SD = 0.00159322, SE = 0.000459922
Internal R/T: mean = 4.0175e-05, SD = 1.14256e-05, SE = 3.29828e-06
Swap R/T: mean = 0.000185383, SD = 7.08558e-05, SE = 2.04543e-05
External: invalid
Internal: $\alpha = 0.382$, $c = 0.2108$, $R^2 = 0.986$
Swap: $\alpha = 0.391$, $c = 0.8338$, $R^2 = 0.995$

